

PENDLETON EASTERN OREGON, OR, USA

WMO: 726880

Lat: 45.698N

Lon: 118.855W

Elev: 1486

StdP: 13.92

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24155

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	8.6	15.7	3.0	6.8	10.2	9.5	9.4	18.3	33.1	44.6	28.9	46.0	4.6	150	0.584

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	30.4	97.1	64.7	93.8	63.7	90.3	62.6	67.0	91.5	65.4	89.4	63.8	86.6	9.5	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.5	77.3	71.4	56.3	71.3	70.9	54.3	66.3	70.4	32.2	91.0	30.9	89.2	29.7	86.5	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.6	24.3	20.3	DB	4.5	103.5	9.0	3.3	-2.0	105.9	-7.3	107.8	-12.4	109.6	-18.9	112.0	(4)
(5)			WB	3.7	70.0	8.7	3.2	-2.6	72.4	-7.7	74.3	-12.5	76.1	-18.9	78.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.5	35.3	38.3	44.9	50.3	58.1	64.9	73.8	72.4	64.0	51.6	41.4	34.4	(6)	
	DBStd	15.41	9.61	8.29	6.46	5.98	6.90	6.65	6.13	5.91	6.63	6.75	8.65	9.47	(7)	
	HDD50	1872	458	334	181	67	9	0	0	0	1	57	276	489	(8)	
	HDD65	5233	920	749	624	441	234	80	6	8	96	416	709	950	(9)	
	CDD50	2799	3	5	22	76	261	448	738	695	422	108	17	4	(10)	
	CDD65	687	0	0	0	1	22	78	279	238	67	2	0	0	(11)	
	CDH74	9953	0	0	1	30	394	1232	4029	3249	973	44	1	0	(12)	
	CDH80	4750	0	0	0	3	118	497	2153	1612	358	9	0	0	(13)	
(14) Wind	WSAvg	8.3	7.0	7.7	9.0	9.6	9.1	9.3	8.7	8.5	8.0	7.8	7.6	7.3	(14)	
(15) Precipitation	PrecAvg	12.4	1.5	1.1	1.3	1.1	1.2	0.9	0.3	0.4	0.6	1.0	1.5	1.6	(15)	
	PrecMax	18.7	3.9	3.0	2.8	2.8	3.2	2.2	1.5	2.6	2.2	2.7	3.8	4.7	(16)	
	PrecMin	6.8	0.4	0.1	0.3	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.2	(17)	
	PrecStd	2.4	0.8	0.6	0.6	0.7	0.8	0.6	0.3	0.6	0.6	0.6	0.7	0.9	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.2	62.3	69.4	77.8	88.5	96.9	102.4	100.4	93.7	79.7	66.2	60.7	(19)	
		MCWB	49.6	50.0	52.4	58.4	62.5	65.9	67.1	65.9	63.5	58.5	54.3	49.6	(20)	
	2%	DB	54.6	56.7	64.0	72.7	83.2	90.5	98.1	96.3	88.1	72.7	61.1	55.4	(21)	
		MCWB	46.3	47.4	50.5	56.0	60.9	63.6	65.2	64.1	61.6	55.7	51.2	47.1	(22)	
	5%	DB	51.0	53.0	60.1	67.7	78.4	85.6	95.1	92.8	84.1	68.0	57.0	51.1	(23)	
		MCWB	44.4	45.4	48.8	53.2	58.7	61.2	63.9	63.2	60.0	53.6	48.9	44.7	(24)	
	10%	DB	47.8	49.8	56.6	63.0	73.3	81.1	91.5	89.1	79.8	63.7	53.3	47.0	(25)	
		MCWB	42.1	43.1	46.9	50.5	56.6	59.3	62.9	62.1	58.3	51.7	46.3	41.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.5	51.8	54.9	60.9	65.3	67.7	69.8	69.1	66.2	60.9	57.1	50.5	(27)	
		MCDB	56.9	59.0	65.2	74.1	83.8	92.2	96.5	94.0	87.8	74.7	64.3	58.3	(28)	
	2%	WB	47.3	48.8	51.9	57.0	62.4	64.8	67.4	66.4	63.5	57.4	52.8	47.5	(29)	
		MCDB	53.2	55.5	60.9	70.4	79.3	87.2	92.5	90.9	83.3	69.7	59.4	53.7	(30)	
	5%	WB	44.8	45.9	50.0	54.2	60.0	62.6	65.6	64.7	61.3	55.1	49.7	44.9	(31)	
		MCDB	50.2	52.2	58.5	66.1	75.0	82.6	91.3	88.5	80.4	65.4	55.6	51.0	(32)	
	10%	WB	42.5	43.7	47.8	51.6	57.9	60.5	63.9	63.1	59.4	52.9	47.2	42.1	(33)	
		MCDB	47.4	49.4	56.0	61.3	71.2	77.7	88.2	86.0	76.8	62.1	52.5	46.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	11.7	14.9	19.3	21.4	23.5	25.6	30.4	29.3	26.6	21.7	14.9	11.6	(35)	
		MCDBR	16.8	18.3	24.4	28.8	31.4	33.3	34.8	34.0	32.7	27.5	18.8	16.1	(36)	
	5% WB	MCWBR	11.0	11.5	13.8	15.2	13.8	12.5	13.1	13.4	14.0	14.3	11.3	10.4	(37)	
		MCWBR	16.4	17.4	22.7	26.8	28.1	30.6	32.3	31.0	29.9	24.6	17.2	15.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.287	0.290	0.304	0.336	0.355	0.353	0.353	0.353	0.338	0.312	0.293	0.285	(40)	
		taud	2.539	2.555	2.533	2.425	2.393	2.424	2.430	2.421	2.445	2.522	2.537	2.523	(41)	
	Ebn at Noon		266	287	297	293	289	289	288	284	279	273	260	253	(42)	
		Edh at Noon	20	24	28	34	36	36	35	34	30	24	20	18	(43)	
(44) All-Sky Solar Radiation	RadAvg		403	713	1115	1640	1988	2275	2446	2064	1527	943	502	326	(44)	
	RadStd		44	87	89	106	150	116	82	115	94	70	42	38	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	-0.99	N/A	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+160	N/A

Nomenclature: See separate page